

BALKHASH, KazakhstanWMO: **357960**Lat: **46.833N**Lon: **74.967E**Elev: **350**StdP: **97.19**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-24.6	-30.8	0.2	-26.5	-28.4	0.3	-24.1	10.4	-10.4	9.4	-11.3	4.1	40	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	31.7	19.2	30.4	19.2	29.1	18.8	22.1	28.6	21.2	27.9	20.4	27.2	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	15.1	27.0	18.6	14.1	26.2	17.6	13.2	25.6	66.8	28.6	63.6	28.0	60.5	27.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	8.0	-29.9	35.5	2.7	2.0	-31.9	36.9	-33.5	38.1	-35.1	39.3	-37.0	40.7		
(5)			-29.8	24.1	2.8	1.2	-31.8	25.0	-33.5	25.7	-35.1	26.4	-37.1	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-13.7	-11.5	-1.9	8.8	16.0	22.0	23.4	21.9	15.4	7.6	-1.3	-9.5	(6)
(7)		DBStd	13.99	6.05	6.18	6.51	4.91	3.76	3.32	2.84	3.24	4.10	4.42	5.69	6.10	(7)
(8)		HDD10.0	2840	736	603	370	79	3	0	0	0	6	99	338	605	(8)
(9)		HDD18.3	4741	994	837	628	287	89	8	2	8	104	333	588	863	(9)
(10)		CDD10.0	1572	0	0	1	42	190	359	417	370	168	25	0	0	(10)
(11)		CDD18.3	433	0	0	0	1	18	118	160	119	17	0	0	0	(11)
(12)		CDH23.3	3846	0	0	0	9	117	1003	1531	1061	125	1	0	0	(12)
(13)		CDH26.7	1051	0	0	0	2	16	259	462	285	27	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.1	4.3	4.4	4.2	4.3	4.2	4.0	4.1	3.9	3.9	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	142	13	10	12	12	14	14	15	8	6	11	15	14	(15)
(16)		PrecMax	248	33	25	30	33	58	89	93	45	75	36	41	37	(16)
(17)		PrecMin	38	0	0	2	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	42	8	6	7	8	13	16	17	10	11	9	11	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	2.3	14.8	24.3	28.6	32.9	34.1	33.2	30.0	21.2	11.9	3.1	(19)
(20)			MCWB	0.5	0.8	7.2	12.9	16.4	19.0	19.1	18.7	16.4	11.4	7.6	1.7	(20)
(21)		2%	DB	-0.7	0.9	11.3	20.5	26.1	30.9	31.9	31.0	26.4	18.6	9.7	1.3	(21)
(22)			MCWB	-1.1	0.0	6.3	11.1	15.7	19.1	19.8	19.0	15.7	10.9	6.8	0.4	(22)
(23)		5%	DB	-2.6	-0.6	8.6	18.2	24.4	29.4	30.6	29.6	24.3	16.7	7.8	0.1	(23)
(24)			MCWB	-3.1	-1.4	5.0	10.5	15.0	18.8	19.8	18.6	15.0	10.3	5.5	-0.7	(24)
(25)		10%	DB	-4.8	-2.2	6.2	16.3	22.9	28.0	29.2	28.2	22.5	14.6	6.1	-1.3	(25)
(26)			MCWB	-5.3	-3.0	3.6	9.7	14.5	18.6	19.4	18.2	14.3	9.3	4.1	-2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	1.4	9.0	14.7	18.9	22.6	23.8	22.9	19.7	14.0	9.4	1.9	(27)
(28)			MCDWB	0.8	1.9	12.7	21.0	25.2	28.8	30.3	29.5	25.7	18.0	10.9	2.5	(28)
(29)		2%	WB	-1.2	0.1	6.7	13.0	17.5	21.4	22.4	21.6	17.7	12.3	7.3	0.6	(29)
(30)			MCDWB	-0.8	0.8	10.3	18.5	23.3	27.9	28.9	28.2	23.6	16.8	8.9	1.1	(30)
(31)		5%	WB	-3.1	-1.3	5.2	11.6	16.5	20.5	21.5	20.5	16.4	11.0	5.7	-0.6	(31)
(32)			MCDWB	-2.7	-0.7	8.3	16.7	22.4	27.0	28.3	27.3	22.5	15.4	7.5	0.1	(32)
(33)		10%	WB	-5.3	-3.0	3.7	10.3	15.5	19.6	20.7	19.6	15.1	9.6	4.2	-2.2	(33)
(34)			MCDWB	-4.9	-2.2	6.1	15.5	21.3	26.3	27.6	26.6	21.6	14.2	6.0	-1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	8.4	8.9	9.9	9.2	9.3	10.1	10.4	9.2	7.6	7.6	(35)
(36)			MCDBR	7.4	8.2	10.3	11.7	11.8	10.7	10.8	11.5	12.2	11.5	9.0	7.1	(36)
(37)			MCWBR	7.3	7.7	6.4	6.1	6.6	6.7	7.0	7.1	7.9	7.7	6.6	6.5	(37)
(38)		5% WB	MCDBR	7.4	7.9	9.6	9.5	9.6	8.9	9.3	9.5	10.3	9.8	7.7	6.5	(38)
(39)			MCWBR	7.3	7.6	6.3	5.7	6.5	6.6	7.0	7.6	8.5	7.6	6.3	6.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.351	0.398	0.442	0.445	0.439	0.449	0.404	0.364	0.366	0.346	0.317	(40)	
(41)		taud	2.026	1.968	2.048	2.034	2.081	2.167	2.156	2.270	2.336	2.233	2.156	2.073	(41)	
(42)		Ebn at Noon	722	780	808	810	823	830	814	840	844	768	692	674	(42)	
(43)		Edh at Noon	100	131	142	159	157	145	145	124	106	100	88	85	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.42	2.31	3.63	5.56	6.89	7.20	6.88	6.38	4.99	3.06	1.67	1.05	(44)	
(45)		RadStd	0.26	0.22	0.39	0.32	0.32	0.39	0.42	0.25	0.23	0.23	0.16	0.20	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page